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AksharaPlay: Low-Cost Dyslexia Screening & Multi-Lingual Game Suite Early primary school diagnostic ecosystem powered by edge speech-recognition and NLP
Theme: Smart Education / Inclusive Tech Category: Software Team ID: [Insert Team ID]

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Proposed Solution A gamified mobile application disguised as an immersive phonics adventure game that screens children for dyslexia during early primary school years.

Core Value: Bypasses clinical psychologist shortages by shifting preliminary diagnostic tracking directly into standard school classrooms. 3-Tiered Architecture 1. Gamified Tracking: Captures touch latency and erratic visual saccades during gameplay. 2.

Speech & NLP: Analyzes phoneme substitution and syllable omission in regional Indian languages. 3. Teacher Dashboard: Converts metrics instantly into actionable clinical risk reports. Idea Title & Proposed Solution

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Frontend & Game Flutter cross-platform mobile & tablet framework built for public school device compatibility. Speech Engine Offline TensorFlow Lite & Vosk API running locally to evaluate regional phonemes without cloud latency. Backend & Analytics Python FastAPI paired with scikit-learn classification models and secure PostgreSQL caching. Technical Approach

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Risk Mitigation Pipeline Technical Feasibility: Built using mature open-source audio frameworks, allowing rapid prototyping and local offline execution in remote rural classrooms. Noise Interference High background noise in crowded rural classrooms. Zero Connectivity Lack of internet in remote primary schools. Child Test Anxiety Intimidation caused by formal diagnostic exams. Feasibility and Viability

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10M+ Undiagnosed Children Saved from mislabeling 90% Cost Reduction Versus private clinics 100% NEP 2020 Aligned Inclusive public schooling Social Inclusion & Emotional Relief Stigma-free early screening prevents rural children from being unfairly mislabeled as slow learners, building lifelong confidence. Democratizing Healthcare Access Completely eliminates heavy financial barriers of private clinical assessments for low-income and marginalized families. Impact and Benefits

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{5C22544A-7EE6-4342-B048-85BDC9FD1C3A}# Reference Framework / Source Core Contribution to Project 1 National Education Policy (NEP) Foundational literacy, numeracy, and inclusive school ecosystem guidelines. 2 Dyslexia Association of India (DAI) Statistical data on uncertified learning disabilities and bottlenecks. 3 Multilingual Speech Standards Acoustic modeling literature for low-resource regional Indian phonemes. 4 Gamified Cognitive Case Studies Frameworks for non-invasive neuro-developmental mobile tracking. Research and References